

Independent Learning with Local Information in Mean-Field Games

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Abstract—This paper studies decentralized reinforcement learning in N -player mean-field games. Our study has two principal objectives: (i) to identify structural properties of games that can guide algorithm design, and (ii) to study the convergence of independent learners under decentralized information. Toward the first objective, we show that N -player mean-field games admit a desirable graph theoretic property. We then study a decentralized learning algorithm in which each agent naively assumes it faces an MDP in its observation variable and periodically revises its policy using an ϵ -satisficing rule: if the agent perceives itself to be ϵ -best-responding to the environment, it does not switch policy; otherwise, the agent incorporates random search of its policy space. We study this learning algorithm under two information structures. When global state information is available to each agent, we establish convergence to near equilibrium. When only local and mean-field state information is available to each agent, we show convergence to policies that can be interpreted as subjective equilibrium.

I. INTRODUCTION

Mean-field games (MFGs) are a relatively recent theoretical framework for studying strategic environments with a large number of weakly coupled decision-making agents [1], [2], [3], [4], [5]. In a MFG, the cost and state dynamics of any particular agent are influenced by the collective behaviour of others only through a distributional *mean-field* term. Given the ubiquity of large-scale decentralized systems in modern engineering, MFGs have been used to model a diverse range of applications, such resource management [6], [7], social conventions [8], power control in telecommunications networks [9], and traffic control [10] among many others.

Mean-field games can be viewed as limit models of N -player symmetric stochastic games, where players are exchangeable and symmetric entities. A number of papers have formally examined the connection between games with finitely many players and the corresponding limit model, including the works of [11], [12], [13].

With the recent empirical successes of single-agent reinforcement learning, e.g. [14], learning agents are increasingly being deployed in a variety of strategic environments, sometimes in the presence of other learning agents. When multiple agents learn, adapt, and change behaviour in a shared environment, the stream of data received by any individual agent becomes non-stationary, and the guarantees of single-agent learning theory fail to carry over. As a result, unexpected and at times undesirable system-wide behaviour

can emerge when incompatible learning agents are used in a shared environment [15].

Multi-agent reinforcement learning (MARL) is the study of the emergent behaviour in systems of interacting learning agents, with stochastic games serving as the most popular framework for modelling such systems [16], [17]. In recent years, there has been a considerable amount of research in MARL that has aimed to produce algorithms with desirable system-wide performance and convergence properties. While these efforts have lead to a number of empirically successful algorithms, such as [18], there are comparatively fewer works that offer formal convergence analyses of their algorithms.

The majority of theoretical contributions in MARL have focused on highly structured classes of stochastic games, such as two-player zero-sum games [19], [20] and N -player stochastic teams and their generalizations [21], [22]. In much of the existing literature on MARL, a great deal of information is assumed to be available to the agents while they learn. These assumptions, such as full state observability ([19]–[22]) or action-sharing among all agents (e.g. [23], [24], [25]), are appropriate in some settings but are unrealistic in many large-scale, decentralized systems modelled by MFGs.

In this paper, we study MARL in N -player, discrete-time MFGs, using the model presented in [12]. We focus on online¹ and decentralized learning: players do not observe the actions of other agents and may have a limited view of the overall state of the system. Furthermore, given that decentralization and learning is central to our study, we focus on algorithms that have minimal coordination between agents after play begins. In particular, we avoid algorithms that require all agents to use the same policy *during learning*, which is a commonly adopted approach, as we will discuss later. In this context, we are interested in developing decentralized algorithms for MFGs that have desirable convergence properties.

This paper makes the following contributions: (1) We show that N -player MFGs have useful graph theoretic structure, which we call the ϵ -satisficing paths property. (2) We design a learning algorithm that exploits this structure and we study its convergence properties under two decentralized information structures. Under a mild synchronization assumption, we show that the game eventually reaches an approximate equilibrium with high probability when the global state is available to each agent. When players use only local state information, we argue that play approaches a policy that can be interpreted as a subjective equilibrium.

This paper is organized as follows. The model for N -

This research was supported in part by the Natural Sciences and Engineering Research Council (NSERC) of Canada.

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¹Players view only the stream of data encountered during gameplay

player mean-field games is stated in Section II along with results on the existence of equilibrium. Section III presents the notion of satisficing paths, along with various related structural results and an illustration of how these concepts can guide the design of independent learning algorithms. Our learning algorithms are presented in Section IV, with §IV-A considering the problem with global state information and §IV-B studying the problem with only mean-field state information. Section V surveys some related literature and compares this paper to other works on learning in MFGs, and the final section concludes.

Notation: We use $\Pr$ to denote a probability measure on some underlying probability space. For finite sets S, S' , we let $\mathcal{P}(S)$ denote the set of probability measures on S , and we let $\mathcal{P}(S'|S)$ denote the set of transition kernels on S' given S . An element $\mathcal{T} \in \mathcal{P}(S'|S)$ is a collection $\{\mathcal{T}(\cdot|s) : s \in S\}$, where for each $s \in S$, $\mathcal{T}(\cdot|s) \in \mathcal{P}(S')$ is a probability measure on S' . We use $Y \sim \mu$ to denote that the random variable Y has distribution μ . For $s \in S$, we use $\delta_s \in \mathcal{P}(S)$ to denote the Dirac measure centered at s .

II. MODEL

A. Stochastic Games

A finite stochastic game with discounted costs is given by

$$\mathcal{G} = (\mathcal{N}, \mathbf{X}, \{\mathbb{U}^i, c^i, \beta^i\}_{i \in \mathcal{N}}, P, \nu_0). \quad (1)$$

The components of $\mathcal{G}$ are such: $\mathcal{N}$ is a finite set of $N \in \mathbb{N}$ players/agents. $\mathbf{X}$ is a finite set of states. For agent $i \in \mathcal{N}$, $\mathbb{U}^i$ is a finite set of actions, and we write $\mathbf{U} := \times_{i \in \mathcal{N}} \mathbb{U}^i$. An element $\mathbf{u} \in \mathbf{U}$ is called a *joint action*. For agent i , $c^i : \mathbf{X} \times \mathbf{U} \rightarrow \mathbb{R}$ is a stage cost function, and $\beta^i \in [0, 1)$ is a discount factor. A random initial state $\mathbf{x}_0 \in \mathbf{X}$ is given by $\mathbf{x}_0 \sim \nu_0$. The transition kernel $P \in \mathcal{P}(\mathbf{X}|\mathbf{X} \times \mathbf{U})$ describes state transition probabilities through equation (2), below.

The playing of $\mathcal{G}$ is summarized as follows. At time $t \in \mathbb{Z}_{\geq 0}$, the state variable is denoted by $\mathbf{x}_t \in \mathbf{X}$, and the action selected by agent $i \in \mathcal{N}$ is denoted by $u_t^i \in \mathbb{U}^i$, while the joint action is denoted by $\mathbf{u}_t = (u_t^j)_{j \in \mathcal{N}} \in \mathbf{U}$. Upon selection of the joint action $\mathbf{u}_t$, each agent $i \in \mathcal{N}$ observes its realized cost $c^i(\mathbf{x}_t, \mathbf{u}_t) \in \mathbb{R}$, and the system transitions to the next state $\mathbf{x}_{t+1}$ in a Markovian fashion according to (2):

$$\Pr(\mathbf{x}_{t+1} = \mathbf{x}' | \mathbf{x}_k = \mathbf{s}_k, \mathbf{u}_k = \mathbf{a}_k, 0 \leq k \leq t) = P(\mathbf{x}' | \mathbf{s}_t, \mathbf{a}_t), \quad \forall \mathbf{x}', \mathbf{s}_0, \dots, \mathbf{s}_t \in \mathbf{X}, \mathbf{a}_0, \dots, \mathbf{a}_t \in \mathbf{U}. \quad (2)$$

For ease of notation, we use boldface characters to denote joint objects, such as $\mathbf{u}_t = (u_t^i)_{i \in \mathcal{N}}$ above. To isolate the role of a particular player $i \in \mathcal{N}$, a joint object with i 's component removed is written with $-i$ in the agent index, e.g. $\mathbf{u}^{-i} = (u^j)_{j \neq i}$.

A *policy* is a rule for selecting actions $(u_t^i)_{t \geq 0}$ using only the information that is locally available at the time of each decision, possibly with randomization. We describe two particular information structures in the next subsection and continue here in the general case. At time $t \geq 0$, player i observes the random quantity h_t^i , which represents the partial history of the system observable by i at time t and

takes values in a set H_t^i . An admissible policy for agent i is a sequence of transition kernels $\gamma^i = (\gamma_t^i)_{t \geq 0}$ such that $\gamma_t^i \in \mathcal{P}(\mathbb{U}^i | H_t^i)$. When agent i follows policy γ^i , the sequence of actions $(u_t^i)_{t \geq 0}$ is then generated according to $u_t^i \sim \gamma_t^i(\cdot | h_t^i)$ for each $t \geq 0$. We denote the set of admissible policies for agent i by Γ^i .

Fixing a joint policy $\gamma \in \Gamma$ induces a probability measure $\Pr^\gamma$ on the set of state-action trajectories $(\mathbf{X} \times \mathbf{U})^\infty$ in a canonical way. Letting E^γ denote the associated expectation, we define the cost criterion for player i as the expectation of its series of discounted costs:

$$J_{\mathbf{x}}^i(\gamma^i, \gamma^{-i}) = E^\gamma \left[\sum_{t \geq 0} (\beta^i)^t c^i(\mathbf{x}_t, u_t^i, \mathbf{u}_t^{-i}) \middle| \mathbf{x}_0 = \mathbf{x} \right] \quad (3)$$

Observe that player i 's cost criterion depends on the joint action sequence $(\mathbf{u}_t)_{t \geq 0}$, but player i only has direct control over its own action sequence $(u_t^i)_{t \geq 0}$. This motivates the following definitions.

Definition 1: Let $i \in \mathcal{N}$, $\epsilon \geq 0$. For $\gamma^{-i} \in \Gamma^{-i}$, a policy $\gamma^{*i} \in \Gamma^i$ is called an ϵ -best-response to γ^{-i} if

$$J_{\mathbf{x}}^i(\gamma^{*i}, \gamma^{-i}) \leq \inf_{\gamma^i \in \Gamma^i} J_{\mathbf{x}}^i(\gamma^i, \gamma^{-i}) + \epsilon, \quad \forall \mathbf{x} \in \mathbf{X}. \quad (4)$$

Definition 2: A joint policy $\gamma^* = (\gamma^{*i})_{i \in \mathcal{N}} \in \Gamma$ constitutes an ϵ -equilibrium if γ^{*i} is an ϵ -best-response to γ^{*-i} for each agent i .

When $\epsilon = 0$ in the definitions above, the terms *best-response* and *equilibrium* are typically used instead of 0-best-response and 0-equilibrium, respectively.

For any $\epsilon \geq 0$, we use $\Gamma_{\text{eq}}^\epsilon$ to denote the set of ϵ -equilibrium policies of the game $\mathcal{G}$.

B. N-Player Mean Field Games

We now introduce our model of N -player mean-field games as a special case of the general stochastic games. The aim is to capture the distinguishing characteristics of a mean-field game—namely, interchangeable, weakly coupled agents—in a model with only finitely many players. The model below is similar to that of [12], which studies existence of equilibrium and allows for general state and actions spaces.

Definition 3: A finite, discounted stochastic game $\mathcal{G}$,

$$\mathcal{G} = (\mathcal{N}, \mathbf{X}, \{\mathbb{U}^i, c^i, \beta^i\}_{i \in \mathcal{N}}, P, \nu_0),$$

is called an N -player mean-field game if the following conditions hold:

- $\mathbf{X} = \times_{j \in \mathcal{N}} X_{\text{loc}}$ for some finite set X_{loc} ;
- For each $\mathbf{x} \in \mathbf{X}$, define $\text{emp}_N(\cdot | \mathbf{x}) \in \mathcal{P}(X_{\text{loc}})$ by

$$\text{emp}_N(\cdot | \mathbf{x}) = \frac{1}{N} \sum_{j \in \mathcal{N}} \delta_{x^j}(\cdot),$$

and let $\text{Emp}_N := \{\text{emp}_N(\cdot | \mathbf{x}) : \mathbf{x} \in \mathbf{X}\}$.

- For all $i, j \in \mathcal{N}$, $\beta^i = \beta^j =: \beta$ and $\mathbb{U}^i = \mathbb{U}^j =: \mathbb{U}$;
- There exists $c : X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}) \times \mathbb{U} \rightarrow \mathbb{R}$ such that for all $i \in \mathcal{N}$, $c^i(\mathbf{x}, \mathbf{u}) = c(x^i, \text{emp}_N(\cdot | \mathbf{x}), u^i)$;

- The state transition kernel $P \in \mathcal{P}(\mathbf{X}|\mathbf{X} \times \mathbf{U})$ has the following product structure:

$$P(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) = \prod_{i \in \mathcal{N}} P_{\text{loc}}(s_{t+1}^i | s_t^i, \text{emp}_N(\cdot|\mathbf{s}_t), a_t^i),$$

for all $\mathbf{s}_{t+1}, \mathbf{s}_t \in \mathbf{X}, \mathbf{a}_t \in \mathbf{U}$, where

$$P_{\text{loc}} \in \mathcal{P}(X_{\text{loc}}|X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}) \times \mathbb{U}).$$

The set X_{loc} above is called the set of local states. Any $\mathbf{x} = (x^i)_{i \in \mathcal{N}} \in \mathbf{X}$ is called a global state, while i 's component $x^i \in X_{\text{loc}}$ is called player i 's local state. The distribution $\text{emp}_N(\cdot|\mathbf{x})$ is the empirical distribution of local states for a particular global state. We refer to each empirical distribution $\text{emp}_N(\cdot|\mathbf{x})$ as a *mean-field state*.

The weak coupling of agents in an N -player MFG can be appreciated by looking at agent i 's cost criterion:

$$\begin{aligned} J_{\mathbf{x}}^i(\gamma) &:= E^\gamma \left[\sum_{t \geq 0} \beta^t c^i(\mathbf{x}_t, \mathbf{u}_t) \middle| \mathbf{x}_0 = \mathbf{x} \right] \\ &= E^\gamma \left[\sum_{t \geq 0} \beta^t c(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t), u_t^i) \middle| \mathbf{x}_0 = \mathbf{x} \right]. \end{aligned}$$

From the second equation above, one observes that agent i 's optimization problem depends on the other players through the evolution of $\{\text{emp}_N(\cdot|\mathbf{x}_t)\}_{t \geq 0}$ and its effect on the evolution of the local state $\{x_t^i\}_{t \geq 0}$.

C. Information, Decentralization, and Structural Results

We now formalize the notion of decentralization by introducing two alternative information structures. The first information structure, of global state and local action information, is typically not assumed in works on MFGs, but is a standard information structure in MARL. The second information structure, of mean-field state and local action information, is often available in large-scale, decentralized systems that can be modelled as MFGs.

Global State and Local Action

Definition 4: For $i \in \mathcal{N}$, define $H_0^i = \mathbf{X}$ and $H_t^i := (\mathbf{X} \times \mathbb{U}^i)^t \times \mathbf{X}$. If $h_0^i = \mathbf{x}_0$ and furthermore the information variables satisfy

$$h_t^i = (\mathbf{x}_0, u_0^i, \mathbf{x}_1, \dots, u_{t-1}^i, \mathbf{x}_t), \quad \forall t \geq 1,$$

then we say player i has *global state and local action information*.

We note that the preceding definition applies to any stochastic game, including games that are not mean-field games. Assuming global state information is rather common in the existing literature on MARL. This is a standard assumption in the literature on *independent learners*; we selectively cite [26], [27], [28], [19], [22], [29] for some examples of MARL with global state and local action information.

Global state observability is difficult to justify in a MFG setting. On one hand, it is perhaps unreasonable to assume knowledge of local state information at faraway agents in

a truly decentralized system. On the other hand, even if such information is available at a particular agent, it may be unwieldy: the size of the global state set is exponential in the number of agents ($|\mathbf{X}| = |X_{\text{loc}}|^N$); moreover, since agents are only weakly coupled, it is conceivable that using global state information offers only negligible performance benefits. These points motivate our second information structure.

Mean-Field State and Local Action

Definition 5: Let $\mathcal{G}$ be an N -player MFG. For $i \in \mathcal{N}$, let $H_0^i := X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}})$ and for $t \geq 1$ let

$$H_t^i := (X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}) \times \mathbb{U})^t \times X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}).$$

If $h_0^i = (x_0^i, \text{emp}_N(\cdot|\mathbf{x}_0))$ and furthermore

$$h_{t+1}^i = (h_t^i, u_t^i, x_{t+1}^i, \text{emp}_N(\cdot|\mathbf{x}_{t+1})), \quad \forall t \geq 0,$$

then we say that player i has *mean-field state and local action information*.

The information structure defined above is standard in the literature on MFGs.

D. Stationary Policies and Equilibrium

As mentioned in §II-A, specifying an information structure determines the set of admissible policies for each agent. For agent $i \in \mathcal{N}$, the set Γ^i of all admissible policies is, however, a cumbersome set to work with in practice, as it allows for policies that are arbitrarily complex and history-dependent. This is especially troublesome when optimizing over Γ^i , as in (4). When justifiable, it is thus desirable to work with restricted subsets of policies within the admissible policies.

Stationary Policies (with global state): The first set of policies we consider is that of *global stationary policies*, under the assumption that player $i \in \mathcal{N}$ has global state information. We denote this set by $\Gamma_G^i \subset \Gamma^i$. The set Γ_G^i is identified with the set $\mathcal{P}(\mathbb{U}^i|\mathbf{X})$ of transition kernels on player i 's actions given the global state. The subscript G is meant to remind the reader that the global state variable is required for such a policy.

When player i follows a global stationary policy $\gamma^i \in \Gamma_G^i$, it selects action $u_t^i \sim \gamma^i(\cdot|\mathbf{x}_t)$. That is, its action is selected randomly in a manner that depends only on the current global state. Under the assumption that all players have global state and local action information, the following is well-known:

- If players $-i$ use a stationary policy $\gamma^{-i} \in \Gamma_G^{-i}$, player i faces a single agent MDP with controlled state process $\{\mathbf{x}_t\}_{t \geq 0}$ and thus it has a best-response (minimizing (4)) within the subset $\Gamma_G^i \subset \Gamma^i$.
- For any $\epsilon \geq 0$, a ϵ -equilibrium policy exists among the stationary Markov policies, i.e. $\Gamma_{\text{eq}}^\epsilon \cap \Gamma_G \neq \emptyset$.

Stationary Policies (with mean-field state): The second class of policies we consider here are those where player i uses only its partial view of the global state when selecting actions, under the assumption that player i has (at least) local state and mean-field state information. We denote the set of *mean-field stationary policies* by Γ_{MF}^i and identify it with the set of transition kernels $\mathcal{P}(\mathbb{U}|X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}))$. Thus, when player i follows a policy $\gamma^i \in \Gamma_{MF}^i$, it selects $u_t^i \sim$

$\gamma^i(\cdot|x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))$, where this distribution depends only on the local state x_t^i and the empirical distribution of local states across all agents, $\text{emp}_N(\cdot|\mathbf{x}_t)$.

We note that since $(x^i, \text{emp}_N(\cdot|\mathbf{x}))$ is a function of the global state $\mathbf{x}$, any mean-field stationary policy is also a global stationary policy. That is, a player i with global state information can implement any policy in Γ_{MF}^i . The reverse is not true: if an agent i has local and mean-field state information as above but does not access the global state, then there are global stationary policies that the player cannot implement.

Unlike in the case with global state information, here it is not always the case that player i always has a best-response to a policy $\gamma^{-i} \in \Gamma_{MF}^{-i}$ within i 's set of mean-field stationary policies Γ_{MF}^i . This is the case because player i does not face an MDP with controlled state process $\{(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))\}_{t \geq 0}$ for general $\gamma^{-i} \in \Gamma_{MF}^{-i}$. For this reason, it is not immediate that $\Gamma_{\text{eq}}^\epsilon \cap \Gamma_{MF}^\epsilon \neq \emptyset$ for some $\epsilon \geq 0$. Indeed, in more general models such as that of [12], a careful argument is required to prove that even approximate equilibrium exists in a larger set of policies.

We conclude this section by proving that for any $\epsilon \geq 0$, $\Gamma_{MF} \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$ in our model of N -player MFGs. The following definition is required.

Definition 6: Let $\mathcal{G}$ be an N -player MFG with mean-field state and local action information at each agent. For players $i, j \in \mathcal{N}$, the policies $\gamma^i \in \Gamma_{MF}^i$ and $\gamma^j \in \Gamma_{MF}^j$ are called *mean-field symmetric* if they are identified with the same transition kernel in $\mathcal{P}(\mathbb{U}|X_{\text{loc}} \times \mathcal{P}(X_{\text{loc}}))$.

Theorem 1: Let $\mathcal{G}$ be an N -player MFG and let $\epsilon \geq 0$. Then, $\Gamma_{MF} \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$.

Proof Sketch: Consider $\gamma^{-i} \in \Gamma_{MF}^{-i}$ such that γ^j and γ^k are mean-field symmetric $\forall j, k \in \mathcal{N} \setminus \{i\}$. By leveraging this symmetry, one can show that $(x_{t+1}^i, \text{emp}_N(\cdot|\mathbf{x}_{t+1}))$ is conditionally independent of $\mathbf{x}_t$ given $(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t), u_t^i)$. Thus, under this symmetry assumption on γ^{-i} , player i faces an MDP with controlled state process $\{(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))\}_{t \geq 0}$. It follows that i has a best-response to γ^{-i} within the set Γ_{MF}^i , and furthermore that the set of such best-responses is compact and convex.

Since players $j, k \neq i$ follow symmetric policies, one may identify each symmetric γ^{-i} with the policy of any agent j , $\gamma^j \in \mathcal{P}(\mathbb{U}|X_{\text{loc}} \times \text{Emp}_N) = \Gamma_{MF}^j$. We then construct a set-valued correspondence $\varphi : \Gamma_{MF}^i \rightarrow 2^{\Gamma_{MF}^i}$ that maps $\gamma \in \Gamma_{MF}^i$ to the set of mean-field stationary best-responses to the symmetric joint policy $\gamma^{-i} := (\gamma)_{j \in \mathcal{N} \setminus \{i\}} \in \Gamma_{MF}^{-i}$.

Note that if $\{\gamma_n^{-i}\}_{n \geq 0}$ is a sequence in Γ_{MF}^{-i} converging to some $\gamma_\infty^{-i} \in \Gamma_{MF}^{-i}$ and $\{\gamma_n^i\}_{n \geq 0}$ is a sequence in Γ_{MF}^i such that (i) $\gamma_n^i \rightarrow \gamma_\infty^i \in \Gamma_{MF}^i$ and (ii) for all n , γ_n^i is a best-response to γ_n^{-i} , then γ_∞^i is a best-response to γ_∞^{-i} . (This can be shown using continuity results such as Lemmas 2 and 3 in Section III-B.) This implies that φ is upper hemicontinuous, and one completes the proof by invoking Kakutani's fixed point theorem, as in [30]. $\diamond$

III. SATISFICING PATHS AND EQUILIBRIUM

In this section, we present the notion of *satisficing paths* and a related property, both of which were introduced in a recent paper [29]. In settings where one can establish the existence of satisficing paths that lead to approximate equilibrium, one can exploit this structure by designing MARL algorithms that drive play to approximate equilibrium along such paths.

Definition 7: A sequence of joint policies $(\gamma_k)_{k \geq 0}$ is called an ϵ -satisficing path if, for every player i and $k \geq 0$, γ_k^i is an ϵ -best-response to $\gamma_k^{-i} \Rightarrow \gamma_{k+1}^i = \gamma_k^i$.

Intuitively, agents may change their policies to any arbitrary policy whenever they are not ϵ -best-responding to the remaining players, but it is required that an agent not change its policy when it is already ϵ -best-responding. We note that this definition is not attached to any learning algorithm or specific policy updating rule; it is simply a property of sequences of joint policies. Specific policy updating rules that halt search when ϵ -best-responding have been studied in several papers, e.g. [31], [32].

Definition 8: Let $\mathcal{G}$ be a stochastic game, $\epsilon \geq 0$, and let $\Gamma' \subseteq \Gamma$ be a subset of joint policies for which $\Gamma' \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$. We say that $\mathcal{G}$ has the ϵ -satisficing paths property within Γ' if the following holds: for any $\gamma_0 \in \Gamma'$ there exists an ϵ -satisficing path $(\gamma_k)_{k \geq 0} \subset \Gamma'$ such that for some finite $K < \infty$ we have $\gamma_K \in \Gamma_{\text{eq}}^\epsilon \cap \Gamma'$.

If a game has the ϵ -satisficing paths property within a particular set, it follows that play can be driven to ϵ -equilibrium by iteratively changing the policies of only those players that are not ϵ -best-responding at a given joint policy. In the absence of such a guarantee, there may exist policies from which ϵ -equilibrium is accessible only along paths that require some players to change policies while they are already ϵ -best-responding. Designing independent learning algorithms that drive play along such paths can be a delicate undertaking, as changing policies while already ϵ -best-responding may result to shifting away from an ϵ -equilibrium in some cases, which can threaten the stability of, and convergence to, equilibria.

Some examples of games and policy subsets for which the ϵ -satisficing paths property holds are the following: (i) stateless ($|\mathbf{X}| = 1$) exact potential games and the set of stationary deterministic policies [33]; (ii) stochastic teams (see [21]) and the set of stationary deterministic policies; (iii) two-player general stochastic games with global state information with the policy set Γ_G [29].

A. Satisficing within the Set of Stationary Policies

In the definition of satisficing paths above, one must determine whether γ_k^i is an ϵ -best-response to γ_k^{-i} . This is, in general, a complicated task, since γ_k^{-i} may be an arbitrarily complicated history-dependent admissible policy. In contrast, when γ_k^{-i} is known to be a global stationary policy, determining whether stationary γ_k^i is an ϵ -best-response can be done using the state-action value function of reinforcement learning theory, which we summarize below.

For any $\pi^{-i} \in \Gamma_G^{-i}$, the optimal Q-factors (also known as the state-action value function) for player i playing against π^{-i} are given by a function $Q_{\pi^{-i}}^* \in \mathbb{R}^{\mathbf{X} \times \mathbb{U}}$, where $Q_{\pi^{-i}}^*$ is the unique solution to the fixed point equation

$$Q^i(\mathbf{x}, u^i) = c(x^i, \text{emp}_N(\cdot|\mathbf{x}), u^i) + \beta \min_{a^i} E^{\pi^i, \pi^{-i}}(Q^i(\mathbf{x}_{t+1}, a^i) | \mathbf{x}_t = \mathbf{x}, u_t^i = u^i),$$

defined for all $(\mathbf{x}, u^i) \in \mathbf{X} \times \mathbb{U}$, where π^i is any policy for player i . One can then check whether a stationary policy π^{*i} is a best-response to the stationary policy π^{-i} by verifying whether

$$J_{\mathbf{x}}^i(\pi^{*i}, \pi^{-i}) \leq \min_{a^i \in \mathbb{U}} Q_{\pi^{-i}}^*(\mathbf{x}, a^i) + \epsilon, \quad \forall \mathbf{x} \in \mathbf{X}. \quad (5)$$

The significance of this formulation is that both the value function $J_{\mathbf{x}}^i$ and the optimal Q-factors $Q_{\pi^{-i}}^*$ can be learned during play. Checking whether one is ϵ -best-responding then amounts to checking whether one's current policy is ϵ -greedy with respect to the Q-factors. In subsequent sections, we will consider what happens when players learn an approximation of these quantities (e.g. player i learns $V_{\mathbf{x}}^i$ instead of $J_{\mathbf{x}}^i$ and $W_{\pi^{-i}}^{*i}$ instead of $Q_{\pi^{-i}}^*$) but uses the same type of comparison to determine whether it is satisfied with its current policy.

We now argue that N -player mean-field games have the ϵ -satisficing paths property within Γ_{MF} for any $\epsilon \geq 0$. The following lemma is needed.

Lemma 1: Let $\mathcal{G}$ be an N -player MFG. For $i, j \in \mathcal{N}$, let $\gamma^i \in \Gamma_{MF}^i$ and $\gamma^j \in \Gamma_{MF}^j$ be mean-field symmetric, and let $\gamma^p \in \Gamma_{MF}^p$ for all $p \notin \{i, j\}$. For any $\epsilon \geq 0$, γ^i is an ϵ -best-response to γ^{-i} if and only if γ^j is an ϵ -best-response to γ^{-j} .

Proof: Suppose the claim is not true: for some $\epsilon \geq 0$ and mean-field symmetric policies γ^i, γ^j , we have that i is ϵ -best-responding to γ^{-i} with γ^i while j is not ϵ -best-responding to γ^{-j} with γ^j . That is, for some $\mathbf{s} \in \mathbf{X}$, we have

$$J_{\mathbf{s}}^j(\gamma^j, \gamma^{-j}) > \epsilon + \inf\{J_{\mathbf{s}}^j(\gamma^{*j}, \gamma^{-j}) : \gamma^{*j} \in \Gamma^j\}.$$

Consider the state $\tilde{\mathbf{s}} \in \mathbf{X}$ that swaps i and j 's local states in $\mathbf{s}$ (i.e. $\tilde{s}^i = s^j$; $\tilde{s}^j = s^i$, and $\tilde{s}^k = s^k \forall k \notin \{i, j\}$). Then, the expectation defining $J_{\mathbf{s}}^i(\gamma^i, \gamma^{-i})$ and $J_{\tilde{\mathbf{s}}}^j(\gamma^j, \gamma^{-j})$ is identical, which means γ^i is not an ϵ -best-response to γ^{-i} , which is a contradiction. $\diamond$

Theorem 2: Let $\mathcal{G}$ be an N -player mean-field game. Then, $\mathcal{G}$ has the ϵ -satisficing paths property within Γ_{MF} for any $\epsilon \geq 0$.

The proof of this theorem parallels that of [29, Theorem 8]. We provide an outline here.

Proof: Fix an initial policy $\gamma_0 \in \Gamma_{MF}$, and let S_0 denote the set of players not ϵ -best-responding at γ_0 . If $S_0 = \emptyset$, then γ_0 is itself an ϵ -equilibrium. If $S_0 = \mathcal{N}$, then no player is ϵ -best-responding at γ_0 and so all players may switch their policies. Put $\gamma_1 = \gamma^* \in \Gamma_{MF} \cap \Gamma_{\text{eq}}^\epsilon$, and we have that (γ_0, γ_1) is an ϵ -satisficing path into ϵ -equilibrium.

If, on the other hand, $1 \leq |S_0| \leq N-1$, some players are ϵ -best-responding while others are not. We fix some player

$j(0) \in \mathcal{N} \setminus S_0$ that is ϵ -best-responding and define a policy γ_1 by $\gamma_1^i = \gamma_0^{j(0)}$ if $i \in S_0$ and $\gamma_1^i = \gamma_0^i$ for $i \notin S_0$, and we define $S_1 := \{j \in \mathcal{N} : \gamma_1^j = \gamma_0^{j(0)}\}$. (Equality here is in the sense of being mean-field symmetric.)

Note that (i) $|S_0| \leq |S_1|$, (ii) all players in S_1 use mean-field symmetric policies. By Lemma 1, either all players in S_1 are ϵ -best-responding at γ_1 or none are.²

We then produce an ϵ -satisficing path $\gamma_1, \dots, \gamma_n$ and a corresponding sequence of sets of players $S_1 \subset \dots \subset S_n$ by growing the cohort S_k of players with matching policy at γ_k , $1 \leq k \leq n$. This is done as follows: if the players in S_k are ϵ -best-responding at γ_k and there exists a player $i \notin S_k$ that is not ϵ -best-responding at γ_k , switch i 's policy to that of the players in S_k to obtain γ_{k+1} and $S_{k+1} = S_k \cup \{i\}$. If no such player exists, then γ_k is an ϵ -equilibrium and an ϵ -satisficing path to ϵ -equilibrium has been found and we stop.

Otherwise, if the players in S_k are not ϵ -best-responding and there exists some player $i \notin S_k$ who is ϵ -best-responding at γ_k , we obtain γ_{k+1} by switching the policy of every player in S_k to γ_k^i and we define $S_{k+1} = S_k \cup \{i\}$. (If no such player i exists, we may again switch the policies of all players from γ_k to an arbitrary policy as in the case where $S_0 = \mathcal{N}$. Thus, we define $\gamma_{k+1} = \gamma^* \in \Gamma_{MF} \cap \Gamma_{\text{eq}}^\epsilon$ and stop.)

We then repeat this process at most N times before either (a) an ϵ -equilibrium is found, or (b) the cohort consists of all players, i.e. $S_k = \mathcal{N}$ for some $k \leq N$. In the former case, we have produced an ϵ -satisficing path into ϵ -equilibrium; in the latter case, we find such a path by appending $\gamma_{k+1} = \gamma^*$, any ϵ -equilibrium in Γ_{MF} . $\diamond$

In fact, the proof of Theorem 2 can be used without modification to prove the more general statement below.

Theorem 3: Let $\epsilon \geq 0$ and suppose $\Pi \subseteq \Gamma_{MF}$ satisfies (i) $\Pi^i = \Pi^j$ for all $i, j \in \mathcal{N}$ and (ii) $\Pi \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$. Then, $\mathcal{G}$ has the ϵ -satisficing paths property within Π .

In what follows, we show how one can use random search and quantization in combination with this result to drive policies to equilibrium along ϵ -satisficing paths. The learning algorithms presented in later sections also build on this idea.

B. Quantization of the Policy Space and its Effect on Paths

We now state well-known continuity results for the cost functionals $\{J_{\mathbf{x}}^i : i \in \mathcal{N}, \mathbf{x} \in \mathbf{X}\}$ and the state-action value functions $\{Q_{\pi^{-i}}^* : i \in \mathcal{N}, \pi^{-i} \in \Gamma_G^{-i}\}$. We then use these results to show that the ϵ -satisficing paths property holds within a finite set of policies obtained by quantizing the set of mean-field stationary policies Γ_{MF} .

We begin by noting that in an N -player MFG, the set of empirical distributions on X_{loc} , which we denoted in §II-B by $\text{Emp}_N := \{\text{emp}_N(\cdot|\mathbf{x}) : \mathbf{x} \in \mathbf{X}\} \subset \mathcal{P}(X_{\text{loc}})$, is finite. We also recall that each mean-field stationary policy for player i is identified with a transition kernel in $\mathcal{P}(\mathbb{U}|X_{\text{loc}} \times \text{Emp}_N)$,

²We caution the reader to note that, unlike S_0 , the set S_1 should not be interpreted as the set of players that are not ϵ -best-responding at γ_1 . Rather, S_1 is the first cohort of players with mean-field symmetric policies.

while each global stationary policy for player i is identified with a transition kernel in $\mathcal{P}(\mathbb{U}|\mathbf{X})$.

We identify $\mathcal{P}(\mathbb{U}|X_{\text{loc}} \times \text{Emp}_N)$ with the Cartesian product of probability simplices over the vertices of $\mathbb{U}$, where the product is indexed by $X_{\text{loc}} \times \text{Emp}_N$. Thus, for $i \in \mathcal{N}$, each policy $\gamma^i \in \Gamma_{MF}^i$ is identified with a point in Euclidean space of finite dimension. We use this to put a metric on Γ_{MF}^i as follows:

$$\|\gamma^i - \tilde{\gamma}^i\|_\infty := \max\{|\gamma^i(a^i|s^i, \nu) - \tilde{\gamma}^i(a^i|s^i, \nu)|\},$$

where the maximum is taken over $a^i \in \mathbb{U}$, $s^i \in X_{\text{loc}}$, and $\nu \in \text{Emp}_N$. An analogous metric is defined on Γ_{MF} , where the maximum is taken over players as well. Similarly, we introduce analogous metrics on Γ_G^i and Γ_G .

The following lemmas can be proved mechanically using standard techniques (see, e.g. [29]), where continuity in each case is defined using the appropriate metric.

Lemma 2: For any $i \in \mathcal{N}$, $\mathbf{x} \in \mathbf{X}$, the cost functional $J_{\mathbf{x}}^i : \Gamma_G \rightarrow \mathbb{R}$ is continuous in the joint policy.

Lemma 3: For any $i \in \mathcal{N}$, $\mathbf{x} \in \mathbf{X}$, $u^i \in \mathbb{U}$, the function $f : \Gamma_G^i \rightarrow \mathbb{R}$ given by $f(\pi^{-i}) = Q_{\pi^{-i}}^{*i}(\mathbf{x}, u^i)$ is continuous.

Since the sets $\{\Gamma_G^i : i \in \mathcal{N}\}$, $\{\Gamma_{MF}^i : i \in \mathcal{N}\}$, Γ_G , and Γ_{MF} are compact, the cost functionals and state-action value functions are also uniformly continuous. Since Γ_{MF} contains an equilibrium, it follows that any sufficiently fine quantization $\Pi \subset \Gamma_{MF}$ contains an ϵ -equilibrium.

Lemma 4: For any $\epsilon > 0$, there exists $\xi > 0$ such that if $\Pi \subset \Gamma_{MF}$ is any quantization of Γ_{MF} with bins of radius $r \leq \xi$, then $\Pi \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$.

Theorem 4: Let $\epsilon \geq 0$ and suppose that Π is quantization of Γ_{MF} such that $\Pi \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$. If $\Pi^i = \Pi^j$ for each $i, j \in \mathcal{N}$, then, the game $\mathcal{G}$ has the ϵ -satisficing paths property within Π .

Proof: The result follows from Lemma 4 and Theorem 3. $\diamond$

Unlike Theorem 2, Theorem 4 guarantees the existence of ϵ -satisficing paths to equilibrium within a *finite* subset of policies. Crucially, this means that players can restrict to searching within a finite set Π rather than an uncountable set Γ_{MF} . Since Π^i is a fine quantization of Γ_{MF}^i , i 's restricting its policy to Π^i entails only a small performance loss.

The second desirable consequence of Theorem 4 for algorithm design is such: since the game has the ϵ -satisficing paths property within Π , play can be driven to ϵ -equilibrium by changing only the policies of those players that are “ ϵ -unsatisfied,” so to speak. We thus obtain a stopping condition, whereby player i can settle on a policy whenever it is ϵ -best-responding. In contrast, the stability of an ϵ -equilibrium may be threatened when an algorithm requires a player to change policies while already ϵ -best-responding.

Remark: Taken together, these points remove the need for *coordinated* search of the joint policy space Π : since ϵ -satisficing paths to equilibrium exist within Π and Π is finite, play can be driven to ϵ -equilibrium even by random policy revision by those players that are not ϵ -best-responding. This

point is useful for designing independent learning algorithms and is formalized by Algorithm 1 and Lemma 5, below.

Algorithm 1: ϵ -satisficing Policy Revision Algorithm

```

1 Set Parameters
2    $e^i \in (0, 1)$ : experimentation probability when not
    $\epsilon$ -best-responding
3    $\Pi^i \subset \Gamma_{MF}^i$ : a subset of policies obtained through
   quantization.
4 Initialize  $\pi_0^i \in \Pi^i$ : initial policy
5 for  $k \geq 0$  ( $k^{\text{th}}$  policy update)
6   if  $J_{\mathbf{x}}^i(\pi_k) \leq \min_{a^i} Q_{\pi_k}^{*i}(\mathbf{x}, a^i) + \epsilon$  for all  $\mathbf{x} \in \mathbf{X}$ 
7     then
8        $\pi_{k+1}^i = \pi_k^i$ 
9     else
10       $\pi_{k+1}^i \sim (1 - e^i)\delta_{\pi_k^i} + e^i \text{Unif}(\Pi^i)$ 
11   Go to  $k + 1$ 

```

Lemma 5: Suppose each agent $i \in \mathcal{N}$ updates its policy sequence $\{\pi_k^i\}_{k \geq 0}$ according to Algorithm 1 and that, for each $k \geq 0$, the policy updates for π_{k+1} are conditionally independent across agents given π_k . If $\Pi \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$ and $\Pi^i = \Pi^j$ for all agents $i, j \in \mathcal{N}$, then

$$\lim_{k \rightarrow \infty} \Pr(\pi_k \in \Pi \cap \Gamma_{\text{eq}}^\epsilon) = 1.$$

Proof: Note that $\{\pi_k\}_{k \geq 0}$ is a time homogenous Markov chain on Π , and that for any $\pi^* \in \Pi \cap \Gamma_{\text{eq}}^\epsilon$, the singleton $\{\pi^*\} \subset \Pi$ is an absorbing set, by Lines 7-8 of Algorithm 1. (This is because the condition in Line 7 is equivalent to ϵ -best-responding.)

By Theorem 4, the game has ϵ -satisficing paths within Π . For any $\pi \in \Pi$, let $L_\pi < \infty$ denote the length of a shortest ϵ -satisficing path within Π that starts at π and terminates at a policy in $\Pi \cap \Gamma_{\text{eq}}^\epsilon$. (Call such a policy π^* and note it depends on π .) We also define $p_\pi > 0$ as the probability the Markov chain follows this path when starting at π , i.e.

$$p_\pi := \Pr(\pi_{L_\pi} = \pi^* | \pi_0 = \pi) > 0, \quad \forall \pi \in \Pi.$$

Define $L := \max\{L_\pi : \pi \in \Pi\}$ and $p := \min\{p_\pi : \pi \in \Pi\}$. For any $m \geq 0$, we have that

$$\Pr\left(\bigcap_{j=1}^m \{\pi_{jL} \notin \Pi \cap \Gamma_{\text{eq}}^\epsilon\}\right) \leq (1 - p)^m.$$

Taking $m \rightarrow \infty$ gives the result, $\diamond$

The advantages of approaching decentralized MARL from an ϵ -satisficing paths point-of-view can be appreciated by studying Algorithm 1. First, there is no need for a specialized policy updating rule (such as inertial best-responding [21] or gradient ascent [19]) that takes into account special structure in the game; random policy updating when not ϵ -best-responding alone can drive play to equilibrium. This underscores the second advantage of this approach: here,

players do not need to perform complex, coordinated search of the joint policy set Π —each player simply asks whether they are ϵ -best-responding or not and then switches its policy accordingly.

We note that, in its present form, Algorithm 1 cannot be implemented in an online MARL problem by independent learners: the joint policy π_k is never observed by any individual agent and quantities $J_{\mathbf{x}}^i(\pi_k)$ and $Q_{\pi_k^{-i}}^*(\mathbf{x}, a^i)$ must be learned during play, which will introduce error. Nevertheless, Algorithm 1 motivates the algorithms of the coming sections. Intuitively, a learning analog of Algorithm 1 can be obtained by replacing the conditional in Line 7 by a proxy question that can be estimated or learned using feedback information during play using a particular policy.

We conclude this section with a remark on Line 10 of Algorithm 1. Here, the choice to update the policy according to $\pi_{k+1}^i \sim (1 - e^i)\delta_{\pi_k^i} + e^i \text{Unif}(\Pi^i)$ was somewhat arbitrary. In particular, the choice to uniformly mix over Π^i with probability $e^i > 0$ is needed to ensure the paths to equilibrium exist in the Markov chain of Lemma 5. The choice to remain with one's old policy with probability $1 - e^i$ was arbitrarily picked for ease of exposition, and can be replaced by any suitable distribution over Π^i according to the taste of the system designer. Some choices may include gradient descent projected back onto Π^i or selecting a best-response to π_k^{-i} within Π^i . Such changes may result in a significant speed-up of convergence to equilibrium when they are well-suited to the underlying game, but the guarantee holds in any case due to the uniform randomization.

IV. LEARNING AND CONVERGENCE RESULTS

A. Learning with Global State

In this section, we present our first independent, online learning algorithm for N -player mean-field games, under the assumption that global state and local action information is available to each agent in the game. This rather demanding informational assumption allows us to give a rather strong convergence guarantee, which is presented in Theorem 5.

At a high level, Algorithm 2 is a direct analog to Algorithm 1, where the cost function and state-action value functions are learned and then used to estimate whether the player is presently ϵ -best-responding.

To state Theorem 5, we now fix $\epsilon > 0$ and make the following assumptions on various objects.

Assumption 1: For $\epsilon > 0$ fixed, the set $\Pi \subset \Gamma_{MF}$ is obtained through quantization and satisfies (i) $\Pi^i = \Pi^j$ for all $i, j \in \mathcal{N}$, and (ii) $\Pi \cap \Gamma_{\text{eq}}^\epsilon \neq \emptyset$.

Next we present two parameter restrictions. The tolerance parameter d^i is taken to be positive to account for noise in the learned estimates, but cannot be too large, otherwise poorly performing policies may be mistaken for ϵ -best-responses. The bound $\bar{d}$ below is defined analogous to $\bar{\delta}$ in [29] and depends on ϵ and Π . We also take ρ^i to be positive so as to encourage the agent to explore all actions in each state; $\bar{\rho}$ upper bounds each ρ^i so that each agent's actual policy does not deviate too much from its baseline policy in the

Algorithm 2: Independent Learning

```

1 Set Parameters
2    $\Pi^i \subset \Gamma_{MF}^i$ : quantized policy set
3    $\mathbb{Q}^i \subset \mathbb{R}^{\mathbf{X} \times \mathbf{U}}$  and  $\mathbb{J}^i \subset \mathbb{R}^{\mathbf{X}}$ : compact sets
4    $\{T_k\}_{k \geq 0}$ : a sequence in  $\mathbb{N}$  of learning phase lengths
5   set  $t_0 = 0$  and  $t_{k+1} = t_k + T_k$  for all  $k \geq 0$ .
6    $\rho^i \in (0, 1)$ : action experimentation probability
7    $e^i \in (0, 1)$ : random policy updating probability
8    $d^i \in (0, \infty)$ : tolerance level for sub-optimality
9    $\{\alpha_n^i\}_{n \geq 0}$ : step sizes such that  $\alpha_n^i \in [0, 1]$ ,
       $\sum_n \alpha_n^i = \infty$ ,  $\sum_n (\alpha_n^i)^2 < \infty$ 
10 Initialize  $\pi_0^i \in \Pi^i$ ,  $Q_0^i \in \mathbb{Q}^i$ ,  $J_0^i \in \mathbb{J}^i$  (all arbitrary)
11 for  $k \geq 0$  ( $k^{\text{th}}$  exploration phase)
12   for  $t = t_k, t_k + 1, \dots, t_{k+1} - 1$ 
13     Select
14        $u_t^i \sim \begin{cases} \pi_k^i(\cdot | x_t^i, \text{emp}_N(\cdot | \mathbf{x}_t)), & \text{w.p. } 1 - \rho^i \\ \text{Unif}(\mathbf{U}), & \text{w.p. } \rho^i \end{cases}$ 
15     Receive  $c_t^i := c(x_t^i, \text{emp}_N(\cdot | \mathbf{x}_t), u_t^i)$ , state  $\mathbf{x}_{t+1}$ 
16     Set  $n_t^i = \# \text{ visits to } (\mathbf{x}_t, u_t^i) \text{ in } [t_k, t]$ 
17     Set  $m_t^i = \# \text{ visits to } \mathbf{x}_t \text{ in } [t_k, t]$ 
18      $Q_{t+1}^i(\mathbf{x}_t, u_t^i) = (1 - \alpha_{n_t^i}^i)Q_t^i(\mathbf{x}_t, u_t^i) + \alpha_{n_t^i}^i [c_t^i + \beta \min_{a^i} Q_t^i(\mathbf{x}_{t+1}, a^i)]$ 
19      $J_{t+1}^i(\mathbf{x}_t) = (1 - \alpha_{m_t^i}^i)J_t^i(\mathbf{x}_t) + \alpha_{m_t^i}^i [c_t^i + \beta J_t^i(\mathbf{x}_t)]$ 
20   if  $J_{t_{k+1}}^i(\mathbf{x}) \leq \min_{a^i} Q_{t_{k+1}}^i(\mathbf{x}, a^i) + \epsilon + d^i \forall \mathbf{x} \in \mathbf{X}$ ,
21     then
22        $\pi_{k+1}^i = \pi_k^i$ 
23   else
24      $\pi_{k+1}^i \sim (1 - e^i)\delta_{\pi_k^i} + e^i \text{Unif}(\Pi^i)$ 

```

k^{th} exploration phase, π_k^i . The quantity $\bar{\rho}$ is analogous to $\bar{\rho}$ in [29] and depends on $\epsilon, \Pi, \bar{d}$, and the collection $\{d^i\}_{i \in \mathcal{N}}$.

Assumption 2: $\forall i \in \mathcal{N}$, $d^i \in (0, \bar{d})$ and $\rho^i \in (0, \bar{\rho})$.

Assumption 3: For all $\mathbf{x}, \mathbf{x}' \in \mathbf{X}$, there exists $m \in \mathbb{N}$ and sequences $\mathbf{s}_0 = \mathbf{x}, \mathbf{s}_1, \dots, \mathbf{s}_m, \mathbf{s}_{m+1} = \mathbf{x}'$ and $\mathbf{a}_0, \dots, \mathbf{a}_m$ such that

$$\prod_{k=0}^m P(\mathbf{s}_{k+1} | \mathbf{s}_k, \mathbf{a}_k) > 0.$$

Theorem 5: Let $\mathcal{G}$ be an N -player mean-field game with global state and local action information at each agent. Let $\epsilon > 0$ and $\Pi \subset \Gamma_{MF}$ satisfy Assumption 1. Suppose all players follow Algorithm 2 and that Assumptions 2 and 3 hold. Then for any $\xi > 0$, there exists $\tilde{T} = \tilde{T}(\xi, \epsilon, \Pi, \{d^i, \rho^i\}_{i \in \mathcal{N}})$ such that if $T_k \geq \tilde{T}$ for all k , then

$$\Pr(\pi_k \in \Pi \cap \Gamma_{\text{eq}}^\epsilon) \geq 1 - \xi,$$

for all sufficiently large k .

The proof of Theorem 5 involves showing that, by allowing learning to proceed for long enough with suitably chosen parameters, Q-learning and value learning results in sufficiently accurate values with arbitrarily high probability. Then, one can approximate the baseline policy process

$\{\pi_k\}_{k \geq 0}$ by the analogous process obtained via Algorithm 1. The details are omitted, as they resemble those of [22], [29].

We conclude this section by noting that Algorithm 2 can only be implemented in its stated form when one assumes (at least) global state and local action information at each agent. The next sections weaken this informational assumption.

B. Learning with Local and Mean-Field State

Subjective Satisficing: We now relax the assumption of global state information and assume only mean-field state and local action information. We address the following question: what happens when each player $i \in \mathcal{N}$ naively approaches its learning problem as though it faces an MDP with controlled state process $\{(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))\}_{t \geq 0}$?

We begin by generalizing the notion of ϵ -best-responding and accordingly generalizing the notion of ϵ -satisficing paths. Instead of requiring that players remain with their old policy when ϵ -best-responding *objectively*, we now allow players to use subjective criteria to estimate whether they are ϵ -best-responding. This is done to account for learning error and partial observation of the global state.

Before, e.g. in (5), a policy's optimality was verified in each global state $\mathbf{x} \in \mathbf{X}$ using the true value function $J_{\mathbf{x}}^i$ and the relevant Q-factors (e.g. $Q_{\pi^{-i}}^{*i}(\mathbf{x}, \cdot)$ when $-i$ used $\pi^{-i} \in \Gamma_G$). Under the decentralized information structure of mean-field state and local action information, players cannot verify any inequality on a global state-by-state basis, and learning the functions $J_{\mathbf{x}}^i, Q_{\pi^{-i}}^{*i}(\mathbf{x}, \cdot)$ is also not directly possible.

To ease the notational burden, let $\mathbb{Y} := X_{\text{loc}} \times \text{Emp}_N$ and for each $i \in \mathcal{N}, t \geq 0$, let $y_t^i := (x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))$. With this notation, we now study what happens when each player $i \in \mathcal{N}$ incorrectly assumes that it faces an MDP with controlled state process $\{y_t^i\}_{t \geq 0}$ and applies a learning algorithm analogous to Algorithm 2.

Definition 9: For each player $i \in \mathcal{N}$ and $y \in \mathbb{Y}$, let $V_y^i : \times_{k=1}^N \mathcal{P}(\mathbb{U}|\mathbb{Y}) \rightarrow \mathbb{R}$ and let $\mathcal{V} := \{V_y^i\}_{i \in \mathcal{N}, y \in \mathbb{Y}}$. For each $i \in \mathcal{N}$ and $\pi^{-i} \in \times_{k=1}^{N-1} \mathcal{P}(\mathbb{U}|\mathbb{Y})$, let $W_{\pi^{-i}}^{*i} : \mathbb{Y} \times \mathbb{U} \rightarrow \mathbb{R}$ and let $\mathcal{W} := \{W_{\pi^{-i}}^{*i}\}_{i \in \mathcal{N}, \pi^{-i} \in \Gamma_{MF}^i}$. A policy $\gamma^i \in \Gamma_{MF}^i$ is called a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -best-response to $\gamma^{-i} \in \Gamma_{MF}^{-i}$ if the following holds:

$$V_y^i(\gamma^i, \gamma^{-i}) \leq \min_{a^i \in \mathbb{U}} W_{\gamma^{-i}}^{*i}(y, a^i) + \epsilon, \quad \forall y \in \mathbb{Y}. \quad (6)$$

Here, the input to any V_y^i is a point $\gamma \in \times_{k=1}^N \mathcal{P}(\mathbb{U}|\mathbb{Y})$ and is interpreted as the mean-field stationary policy $(\gamma^i, \gamma^{-i}) \in \Gamma_{MF}^i \times \Gamma_{MF}^{-i}$. A similar remark holds for the subscript γ^{-i} in $W_{\gamma^{-i}}^{*i}$.

Using this subjective criteria for ϵ -best-responding, we obtain subjective analogs to ϵ -equilibrium, ϵ -satisficing paths, and the ϵ -satisficing paths property within a set $\Gamma' \subseteq \Gamma$.

The collection $\mathcal{V}$ is called symmetric if $V_y^i = V_y^j$ for all $i, j \in \mathcal{N}$ and $y \in \mathbb{Y}$. The collection $\mathcal{W}$ is called symmetric if $W_{\pi^{-\text{ind}}}^{*i} = W_{\pi^{-\text{ind}}}^{*j}$ for any $i, j \in \mathcal{N}$ and joint policy $\pi^{-\text{ind}}$.

Lemma 6: Suppose $\mathcal{V}$ and $\mathcal{W}$ are symmetric. Suppose that $\Pi \subset \Gamma_{MF}$ contains at least one $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium and that $\Pi^i = \Pi^j$ for all $i, j \in \mathcal{N}$. Then, $\mathcal{G}$ has the $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -satisficing paths property within Π .

Proof: The argument used to prove Theorem 2 can be used here, too. $\diamond$

The nature and significance of the collections $\mathcal{V}$ and $\mathcal{W}$ —including existence of $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium—is addressed in the next section, where we present a local-information analog to Algorithm 2. Intuitively, the functions $\{V_y^i\}_{y \in \mathbb{Y}}$ are meant to be analogs to the cost functionals $\{J_{\mathbf{x}}^i\}_{\mathbf{x} \in \mathbf{X}}$ and the functions $\{W_{\pi^{-i}}^{*i}\}_{\pi^{-i} \in \Gamma_{MF}^{-i}}$ are meant to be analogs to the Q-functions $\{Q_{\pi^{-i}}^{*i}\}_{\pi^{-i} \in \Gamma_{MF}^{-i}}$. The following lemmas say that when continuity properties analogous to Lemmas 2 and 3 hold, then one can obtain results analogous to Lemma 4 and Theorem 4.

Lemma 7: Let $\mathcal{V}$ and $\mathcal{W}$ be given above. For each $i \in \mathcal{N}$, $y \in \mathbb{Y}$, and $a \in \mathbb{U}$, suppose that V_y^i and $W_{(\cdot)}^{*i}(y, a) : \Gamma_{MF}^{-i} \rightarrow \mathbb{R}$ are continuous. Let $\epsilon > 0$ and assume Γ_{MF} contains a $(\mathcal{V}, \mathcal{W})$ -subjective $\epsilon/2$ -equilibrium. There exists $\xi > 0$ such that if $\Pi \subset \Gamma_{MF}$ is any quantization of Γ_{MF} with bins of radius $r \leq \xi$, then Π contains a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium.

Lemma 8: Suppose $\mathcal{V}$ and $\mathcal{W}$ are symmetric and satisfy the continuity conditions of Lemma 7. Let $\epsilon > 0$ and suppose $\Pi \subset \Gamma_{MF}$ is a quantization of Γ_{MF} containing a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium. If $\Pi^i = \Pi^j \forall i, j \in \mathcal{N}$, then $\mathcal{G}$ has the $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -satisficing paths property within Π .

Learning with Mean-Field State: In this section, we present an algorithm suitable for the decentralized information structure of local and mean-field state and local action information. In essence, the algorithm of this section instructs the agents to naively implement Algorithm 2 but replace $\mathbf{x}_t$ with $y_t^i = (x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))$.

As previously noted, when players $-i$ follow a (global or mean-field) stationary policy π^{-i} , player i faces a MDP with controlled state process $\{\mathbf{x}_t\}_{t \geq 0}$. If player i has global state information, it may implement Algorithm 2 using the process $\{\mathbf{x}_t\}_{t \geq 0}$ and estimate whether its current algorithm is an ϵ -best-response. When i has only local and mean-field state or otherwise chooses to ignore the global state (e.g. to save computational resources), it faces a POMDP with observation variable $y_t^i = (x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))$.

If player i performs Q-learning using only the process $\{y_t^i\}_{t \geq 0}$, this can also be viewed as Q-learning with quantization [34], [35]. It was recently shown that, under mild conditions, Q-learning with quantization converges with probability one to the solution of a fixed point equation [34]. A similar convergence analysis holds for estimates of the value function.

Consider the k^{th} exploration phase of Algorithm 3. Here, each agent i has a baseline policy π_k^i and uses an actual policy $\hat{\pi}_k^i$ that mixes between π_k^i and uniform action selection with probabilities $1 - \rho$ and ρ , respectively. Player i learns against an environment determined by π_k^{-i} . Supposing this exploration phase never ends (i.e. $T_k = \infty$), the results of [34] show that both $\lim_t J_t^i(y)$ and $\lim_t Q_t^i(y, a)$ exist with probability one. We define their limits as $\lim_t J_t^i(y) =: V_y^i(\pi_k)$ and $\lim_t Q_t^i(y, a) =: W_{\pi_k^{-i}}^{*i}(y, a)$. Here, we avoid

an explicit expression for $V_y^i(\pi_k)$ and $W_{\pi_k^i}^{*i}$ for brevity of exposition, but note that they are obtained using a fixed-point equation as done in [34].

In the special case that π_k^j and π_k^p are mean-field symmetric for all $j, p \neq i$, we have that i faces an MDP with controlled state process $\{(x_t^i, \text{emp}_N(\cdot|\mathbf{x}_t))\}_{t \geq 0}$. In this case, the objects $\{V_y^i(\pi_k^i)\}_{y \in \mathbb{Y}}$ are actually the value functions for the MDP faced by i and the function $W_{\pi_k^i}^{*i}$ is the state-action value function. Thus, in this case, i can *objectively* determine whether π_k^i is an ϵ -best-response to π_k^i by checking whether it is a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -best-response.

Indeed, as Algorithm 3 requires each agent to use matching parameters, the collections $\mathcal{V}$ and $\mathcal{W}$ (as defined above) are symmetric. It follows that any symmetric ϵ -equilibrium $\gamma_\epsilon^* \in \Gamma_{MF}$ is also a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium. By Theorem 1, there exists a symmetric 0-equilibrium in Γ_{MF} . Then, if $\Pi \subset \Gamma_{MF}$ is a sufficiently fine quantization such that $\Pi^i = \Pi^j$ for all $i, j \in \mathcal{N}$, then Π contains a symmetric ϵ -equilibrium (and thus a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium).

The preceding remarks show that, for a suitably fine symmetric quantization $\Pi \subset \Gamma_{MF}$, an N -player mean-field game $\mathcal{G}$ has the $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -satisficing paths property within Π . Here, $\mathcal{V}$ and $\mathcal{W}$ are collections that have intrinsic meaning, as each function is the limit of a learning process. The $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium joint policies can thus be interpreted as those policies in which a naive learning agent believes itself to be ϵ -best-responding due to the learning it performs.

All together, this sets the stage for Algorithm 3: by using a direct analog of Algorithm 2 that replaces the local state $\mathbf{x}_t$ by the local information y_t^i and using the same mechanics of ϵ -satisficing and random search, we obtain high probability convergence to $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium. To formalize this, we fix $\epsilon > 0$ and make the following assumptions.

Assumption 4: For $\epsilon > 0$ fixed, the set $\Pi \subset \Gamma_{MF}$ is obtained through quantization and satisfies (i) $\Pi^i = \Pi^j$ for all $i, j \in \mathcal{N}$, and (ii) Π contains a $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium.

The following parameter restrictions is analogous to Assumption 2, but we further require that all players use matching parameters. The quantities $\hat{d}$ and $\hat{\rho}$ are defined analogously to $\bar{d}$ and $\bar{\rho}$ in Section ??.

Assumption 5: For all $i \in \mathcal{N}$, $d^i = d \in (0, \hat{d})$ and $\rho^i = \rho \in (0, \hat{\rho})$.

Theorem 6: Let $\mathcal{G}$ be an N -player MFG with mean-field state and local action information at each agent. Let $\epsilon > 0$ and $\Pi \subset \Gamma_{MF}$ satisfy Assumption 4. Suppose all players use Algorithm 3 and Assumptions 3 and 5 hold. For any $\xi > 0$, $\exists \tilde{T} = \tilde{T}(\xi, \epsilon, \Pi, d, \rho)$ such that if $T_k \geq \tilde{T} \forall k$, then

$$\Pr(\pi_k \in \text{Subj}_\epsilon) \geq 1 - \xi,$$

for all sufficiently large k , where Subj_ϵ is the set of $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium policies in Π .

Note: although symmetric equilibrium are guaranteed to exist and thus the set of $(\mathcal{V}, \mathcal{W})$ -subjective ϵ -equilibrium is

guaranteed to be non-empty, it is possible that some *non-equilibrium* mean-field policies also belong to this set of subjective equilibrium. Our convergence claims do not rule out such subjective equilibria.

Algorithm 3: Learning with Local Information

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1 Set Parameters
2    $\Pi^i \subset \Gamma_{MF}^i$ : quantized policy set
3    $\mathbb{Q}^i \subset \mathbb{R}^{\mathbb{Y} \times \mathbb{U}}$  and  $\mathbb{J}^i \subset \mathbb{R}^{\mathbb{Y}}$ : compact sets
4    $\{T_k\}_{k \geq 0}$ : a sequence in  $\mathbb{N}$  of learning phase lengths
5   set  $t_0 = 0$  and  $t_{k+1} = t_k + T_k$  for all  $k \geq 0$ .
6    $\rho^i \in (0, 1)$ : action experimentation probability
7    $e^i \in (0, 1)$ : random policy updating probability
8    $d^i \in (0, \infty)$ : tolerance level for sub-optimality
9    $\{\alpha_n^i\}_{n \geq 0}$ : step sizes such that  $\alpha_n^i \in [0, 1]$ ,
       $\sum_n \alpha_n^i = \infty$ ,  $\sum_n (\alpha_n^i)^2 < \infty$ 
10 Initialize  $\pi_0^i \in \Pi^i$ ,  $Q_0^i \in \mathbb{Q}^i$ ,  $J_0^i \in \mathbb{J}^i$  (all arbitrary)
11 Receive  $y_0^i := (x_0^i, \text{emp}_N(\cdot|\mathbf{x}_0))$ 
12 for  $k \geq 0$  ( $k^{\text{th}}$  exploration phase)
13   for  $t = t_k, t_k + 1, \dots, t_{k+1} - 1$ 
14     Select  $u_t^i \sim \begin{cases} \pi_k^i(\cdot|y_t^i), & \text{w.p. } 1 - \rho^i \\ u^i \sim \text{Unif}(\mathbb{U}^i), & \text{w.p. } \rho^i \end{cases}$ 
15     Receive cost  $c(y_t^i, u_t^i)$ ,  $y_{t+1}^i$ 
16     Set  $n_t^i = \# \text{ visits to } (y_t^i, u_t^i) \text{ in } [t_k, t]$ 
17     Set  $m_t^i = \# \text{ visits to } y_t^i \text{ in } [t_k, t]$ 
18      $Q_{t+1}^i(y_t^i, u_t^i) = (1 - \alpha_{n_t^i}^i) Q_t^i(y_t^i, u_t^i) +$ 
        $\alpha_{n_t^i}^i [c(y_t^i, u_t^i) + \beta \min_{a^i} Q_t^i(y_{t+1}^i, a^i)]$ 
19      $J_{t+1}^i(y_t^i) =$ 
        $(1 - \alpha_{m_t^i}^i) J_t^i(y_t^i) + \alpha_{m_t^i}^i [c(y_t^i, u_t^i) + \beta J_t^i(y_{t+1}^i)]$ 
20   if  $J_{t_{k+1}}^i(y) \leq \min_{a^i} Q_{t_{k+1}}^i(y, a^i) + \epsilon + d^i \quad \forall y \in \mathbb{Y}$ ,
       then
21      $\pi_{k+1}^i = \pi_k^i$ 
22   else
23      $\pi_{k+1}^i \sim (1 - e^i) \delta_{\pi_k^i} + e^i \text{Unif}(\Pi^i)$ 

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V. RELATED WORK AND DISCUSSION

Learning in MFGs is a very active but also very young research area. As a result, there is some inconsistency in naming conventions for discussing the problem under consideration. We now briefly survey literature related to this paper and attempt to clarify some of these inconsistencies.

Early contributions to learning in MFGs include [36] and [37], which studied learning in specific classes of MFGs. Another (relatively) early contribution is that of [38], which does not study MFGs but does is closely related and uses ideas inspired by the mean-field theory of physics.

More related to the present paper are the works [39], [40], [41], [42], [43] and [44]. By and large, these works approach learning in MFGs by analyzing the single-agent problem for a representative agent in the limiting case as $N \rightarrow \infty$, and equilibrium is defined using a best-responding condition

as well as a consistency condition; see, for instance, [40, Definition 2.1]. This notion of equilibrium is inherently symmetric: at equilibrium, all agents use the policy of the representative agent. In contrast, the notion of equilibrium we consider is more along the lines of [40, Definition 5.1], as it allows for different agents to follow different policies.

In [39], the authors study a model of MFGs that allows for costs and state dynamics to depend also on the distribution of actions of other players. Under somewhat stringent assumptions of Lipschitz continuity, an existence and uniqueness of equilibrium result is given, and an algorithm is presented for learning such an equilibrium. Of note, this paper assumes that the learner has access to a population simulator for sampling state transition and cost data at one's convenience. As such, this algorithm is not suitable for online learning applications, where the data observed by the learner depends on the trajectory of online play.

A special case of MFGs, called stationary MFGs, is studied in [40]. The authors present multiple notions of equilibrium and study two-timescale policy gradient methods in this setting, wherein a representative agent updates its policy to best-respond to its (estimated) environment on the fast timescale and updates its estimate for the mean-field flow on the slow timescale. By assuming access to a simulator for obtaining data, convergence to a weak notion of local equilibrium is proved.

MFGs with a finite time horizon and uncountable state and action spaces are considered in [41]. A fictitious play algorithm of sorts is proposed and analyzed, though the question of learning best-response is black-boxed.

Another fictitious play algorithm is proposed in [42], which in some ways parallels [40] by iteratively updating both the policy and the mean-field term. The algorithm of [42] is incorrectly called “online” in reference to its every-step updating of both the policy and the mean-field term. This term is used to highlight that there is no outer loop structure in the algorithm, whereby the optimal policy is estimated iteratively while the mean-field term is held fixed. We note that the algorithm of [42] is not online in the sense of using only data observed during play, since running Algorithm 1 of that work requires the agent to “do nothing” if it so chooses (C.f. Line 4, [42, Algorithm 1]).

Unlike the preceding works, our focus is on online learners that can be employed in a truly decentralized manner. In particular, we avoid assuming either that a simulator is available or that each agent follows the same policy during the learning process. To achieve this, we have imposed an assumption on synchrony in learning: the sequence of exploration phase lengths $\{T_k\}_{k \geq 0}$ is common to all agents in both Algorithms 2 and 3. We view this as a small concession to make in order to allow for the use of differing policies during the learning process.

VI. CONCLUSIONS

In this paper, we studied N -player mean-field games under two different decentralized information structures. We presented the notion of ϵ -satisficing paths and proved that

N -player MFGs possess the ϵ -satisficing paths property for all $\epsilon \geq 0$ within the set of mean-field stationary policies. Building on this structural finding, we proposed learning algorithms that search the set of joint policies in an independent and uncoordinated manner, according to an ϵ -satisficing rule. These algorithms are suitable for application in a decentralized, online learning environment, and we showed that under such algorithms, play converges to approximate equilibrium.

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